

Ground Cargo and Passenger Vehicles

Light/Medium Tactical Trucks: The U.S. Army's workhorse medium truck is the 2.5-ton **M1078 LMTV** (4x4). It "is designed to transport cargo and soldiers" and carries **5,000 lb** of cargo ¹ ². It is powered by a 225 hp Caterpillar 6.6 L diesel engine (running on Diesel, JP-4/JP-8) ³, yielding a top speed of ~58 mph and ~400+ mi range at gross weight ⁴. The truck measures roughly 253" long, 96" wide and 105.5" high (road mode) ⁵. Fold-down sides and optional bench seats allow troop transport ¹; an optional 1,500 lb loader crane (MHE) can load/unload cargo. Standard fuel consumption is roughly 10–12 mpg highway (varies with load). The M1078 bed can carry two 463L pallets side by side (with cargo nets), or standard 40×48" pallet loads. Military cargo is normally organized on unit loads (pallet + net). A unit load standard (MIL-STD-1660) specifies using **40×48"** wood pallets (MIL-P-15011) for Army ground shipments ⁶ (rather than full-size 463L pallets). A 463L Master Pallet (88"×108") is primarily for airlift; it weighs 290 lb and holds up to 10,000 lb ⁷ and can be carried by large trucks or trailers if needed. Typical cargo on LMTVs includes boxed supplies, ammunition, fuel drums, or troops. LMTVs travel in motor pools and on highways, often in platoon convoys (marked and escorted per regulations below).



Figure: An Oshkosh M1078 LMTV (4×4, 2.5 ton) entering a landing craft. The LMTV's fold-down sides and optional troop benches make it a versatile cargo/pax carrier ¹ ⁵.

Heavy Tactical Trucks (HEMTT & PLS): The **Oshkosh HEMTT** (Heavy Expanded Mobility Tactical Truck) family provides much larger capacity (8×8 or 10×10). The HEMTT series (M977 cargo truck, M978 tanker, M1120 LHS load-handler, M984 wrecker, etc.) "provides heavy transport capabilities for supply and re-supply of combat vehicles and weapons systems" ⁸. Compared to earlier 5-ton trucks it offers greatly "**increased payload and mobility**" ⁸. For example, the M977A4 cargo HEMTT has a 10ton rated payload and the M978A4 fuel tanker carries up to 9,500 L of fuel. The M984 HEMTT wrecker (10ton recovery truck) has a

11,000 lb winch and heavy recovery boom ⁹. Loaded HEMTTs weigh ~20–25 tons and typically get ~3–4 mpg on road. Dimensions are roughly 30' long and 8' wide. HEMTTs are diesel-powered (450 hp Detroit Diesel or Caterpillar engines) and run on JP-8/Diesel. They routinely carry ammunition, fuel (bulk and containerized), spare parts, and even howitzers or light armored vehicles. Many HEMTTs tow trailers (5-ton semitrailers or M870 cargo trailers). The **Palletized Load System (PLS)** trucks (M1075/1084) have 10 ton chassis and use 20' flatracks to carry cargo containers or bundled supplies.



Figure: Oshkosh HEMTT M984 wrecker (8×8, 10 ton recovery truck). HEMTT variants (cargo, tanker, tractor, wrecker) all share the same chassis and heavy-duty drivetrain ⁸. The M984's 11,000 lb winch and recovery gear can pull disabled vehicles onto its bed ⁹.

Heavy Equipment Transport (HET): For very heavy loads (tanks, armored vehicles), the Army uses the **Oshkosh M1070 HET** tractor (6×6) with a DRS M1000 semi-trailer. The M1070 is "**the U.S. Army's current tank transporter,**" hauling M1 Abrams tanks and other heavy systems ¹⁰. It has a 500–700 hp diesel engine and can tow ~100 tons. Its dimensions are ~30' tractor + ~70' trailer when loaded; height ~12' (with cargo) and width ~11' (with camo netting). Payload is essentially whatever the M1000 trailer can carry (M1 tank, M88 ARV, etc.). The M1070E1 (newer version) even accommodates heavier payload with higher horsepower. Typically these run at 40–50 mph on highway, ~2–3 mpg under load. Smaller heavy loads (tracked APCs, light armor) may be hauled on semi-trailers by 8×6 tractors (e.g. M915 series).



Figure: M1070 Heavy Equipment Transporter tractor pulling an M1000 trailer (load: M60 Patton tank). The M1070 HET is used to move main battle tanks and other heavy combat vehicles ¹⁰. Its DRS M1000 trailer serves as a mobile flatbed for loads up to ~70–80 tons.

Medium Tactical Vehicles (MTVR, 5-Ton): The Marine Corps' standard medium truck is the **Oshkosh MTVR** (Medium Tactical Vehicle Replacement). It is a 6×6 truck family with a 7-ton (off-road) and 15-ton (road) payload ¹¹ ¹². Variants (MK23–MK38) include cargo (with/without winch), dump, tractor, and wrecker versions. MTVRs are similar to HEMTT in many aspects. “MTVR – standard cargo truck with winch, 184” WB” etc. They use a Caterpillar 425 hp diesel and have comparable fuel use (diesel/JP-8). They typically haul heavy logistics loads and can cross country at ~25 mph. The Army's older **5-ton trucks** (M809/M939 series) serve as tractor units and cargo carriers (payload ~10,000 lb) on a 6×6 chassis; these are being replaced by FMTV trucks in service.

Utility and Personnel Vehicles: Light utility vehicles include the Humvee (HMMWV) and its up-armored variants (M1151 ambulances, M1165 expanded 4-door troop carriers, etc.). These carry ~3–4 soldiers or a small cargo load (~2,500 lb) and typically do ~60 mph on road. Larger buses (e.g. 50-60 passenger military coach buses) are used on bases or in convoys for troop movement. All vehicles follow state traffic laws on public roads (speed limits, weight limits, etc.), plus military rules. For example, overweight or oversized military loads **require escort and clearance** (discussed below).

Military Rail Transport

The U.S. military uses rail extensively for cargo movements. The Department of Defense has designated ~30,000 miles of U.S. railroad corridors as the **Strategic Rail Corridor Network (STRACNET)**, “as essential to the national defense” ¹³. This network links nearly all major ports, depots and bases. Some military installations maintain their own railroads or spurs for internal use and intermodal transfer. For instance, **Fort Eustis, VA** (Army Transportation School) has a 31-mile military railroad with multiple yards and spurs for training and shipment ¹⁴. On the East Coast, **Naval Weapons Station (NWS) Earle, NJ** has two sites connected by “Normandy Road,” a **15-mile military road/rail line** to carry ammunition from inland to the

waterfront ¹⁵ . In California, **Marine Corps Logistics Base Barstow** has an on-base rail yard for logistics support. In Georgia, **MCLB Albany** has railroad tracks crossing the base; in 2013 it reported shipping Marine retrograde equipment by train (“as many as 50 or as few as 30 railcars” used) ¹⁶ . On-base rail stations (“railheads”) serve as transfer points; for example, Fort Eustis RR station at Newport News (Fort Eustis, 37.1569°N, 76.5970°W) and NWS Earle’s Mainside/Waterfront sites (40.2500°N, 74.1500°W and 40.4083°N, 74.0750°W) connect to commercial lines ¹⁴ ¹⁵ . Altogether, the military owns or uses thousands of miles of track (including STRACNET, base rails, and Army/MC reserve rail units) to move heavy cargo (tanks, ammunition, fuel, vehicles) over long distances.

Transport Regulations & Protocols

Military land transport follows detailed doctrine and civilian law. Vehicles must comply with traffic laws on public roads (e.g. weight/width limits, licensing). Special procedures govern military convoys: commanders must obtain movement clearances and mark all vehicles. For example, the Army’s Transportation SOP mandates **convoy clearance** procedures and **“restrictions on tracked, overweight, or outsize vehicles.”** ¹⁷ . Unit commanders use checklists asking, *“Have convoy clearances been obtained? ... Is escort required?”* to ensure proper approval ¹⁸ . Oversize loads (wider than ~11½’ or heavier than state limits) generally need pilot cars and state permits. Tanks and other tracked vehicles are normally not road-legal under their own power; they must be transported on trailers (e.g. M1070/M1000) or moved in specially authorized convoys. Hazardous cargo (fuel, explosives) requires placarding and often armed escort. In short, **all vehicles operating in convoy must be planned with clearances, marked with convoy flags/numbers, and escorted or guided as directed by FM 55-15** ¹⁷ ¹⁸ . If no escort or special requirements are needed, that is noted in the movement clearance.

Cargo Palletization and Loading

Cargo is organized into **unit loads** secured on pallets and nets per military standards. The standard air/land pallet is the 463L (HCU-6/E) pallet (88”×108”). It is used with a lightweight net system for bundling cargo ¹⁹ ⁷ . The 463L pallet itself **“is the main air-cargo pallet of the United States...”**, measuring 108”×88” and supporting up to 10,000 lb ¹⁹ ⁷ . It was designed to work on roller conveyors and cargo handling systems (including ground vehicles) ²⁰ . On land, unit loads usually use **standard pallets (~40”×48”)** to match forklifts and trucks ⁶ . MIL-STD-1660 (ammunition unit loads) explicitly directs that unit loads use MIL-P-15011 wood pallets (40×48”) or similar, with battens and strapping so the load footprint is flush with the pallet ⁶ . Cargo (ammo boxes, MREs, parts) is stacked on these pallets and wrapped with banding or tied with nets. Equipment loads (vehicles, howitzers) are driven onto flatracks or trailers and blocked. In all cases, logistics planners calculate needed vehicles by dividing total cargo weight (and volume) by each vehicle’s capacity. For example, 50,000 lb of supplies would require at least 10 LMTVs (5,000 lb payload each), or fewer heavy trucks (HEMTTs or PLS) if loading heavier loads per truck. Planners also ensure proper center of gravity, clearances, and tie-downs per TM/DDI loading standards.

Sources: U.S. Army technical manuals and fact sheets; DOD transportation regulations; military logistics references ¹ ⁸ ¹¹ ¹³ ¹⁴ ¹⁵ ¹⁷ ¹⁸ ¹⁹ ⁶ ¹⁶ . (For brevity, only key excerpts are cited here; full specs and doctrine are found in the referenced manuals and publications.)

1 2 3 4 5

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